BANKHAUL · FEASIBILITY

BankHaul feasibility assessment

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DOCUMENT CONTROL

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Project TRL	TRL 3 Proof of concept		
Repository	github.com/BoujeeEnjinia1701/bankhaul		
Source	docs/08-feasibility.md @ 659e1df		

REVISION HISTORY

Version	Date	Author	Change
0.2	2026-10-25	Amish Chadha	Station USD 399 with the bank-edge board; O-6 decided (2026-10-25)
0.1	2026-10-25	Amish Chadha	First assessment, closed at TRL 3 from BKH-CAL-001 v0.3, the STEM check and the D3 model (kit 1.8 upgrade)

BankHaul feasibility assessment

Closed at TRL 3: feasible. The calculations (BKH-CAL-001) show the loop sets and hauls a 30 m gillnet at about 200 N with the bank station and far stake at a factor of about 2 in soft ground, and the D3 model shows every part fits (`cad/fidelity.json`). What remains unproven is the ground, the people and the season, not the physics: the bed and bank strength at a real site, the pull the users can give, the weak link's scatter, driving the stake, and UV and bush wear (BKH-STM-001 S4, S6, S17, S19, S22). Method: STANDARDS section 23.

Table 1. Feasibility by aspect.

Aspect	Assessment	Rating	Risks and notes	Evidence
Design	A slack floating loop on two blocks; peak pull about 200 N; far end at most 1.4 kN through the weak links; bank 3 kN	High	Factors near 2 on the soil (anchors 2.13, stake 2.03 in very soft mud); a softer bed needs a new check	BKH-STM-001 S2 to S16; BKH-CAL-001 B to E
Materials	Painted steel tube and plate, galvanised rigging, polyester and polysteel rope, timber	Medium	UV loss of the loop rope over a season; corrosion of the stake at the waterline; grades to confirm	BKH-STM-001 S19, S23; BKH-DEC-001 items 5, 11
Manufacturing and assembly	Saw, drill, MIG or stick welder for 3 to 8 mm steel, carpentry and eye splicing; 11 assembly steps	High	Squareness of the post and the stake's straightness; splicing skill	BKH-BLD-001; PR-013
Power and controls	None; hand-worked	High	None	
Supply of parts	Tube, plate and timber locally; blocks, shackles, turnbuckles, anchors and rope from rigging and fencing suppliers; USD 399 in parts	Medium	Rated rigging may be scarce near the sites; the block price may be low; kept as indicative and to confirm with a local rigging supplier (BKH-DEC-001 O-6, decided 2026-10-25)	bom/bom.csv; BKH-MKT-001
Site and installation	A bank of soft wet soil 6 m or more deep from the edge; a bed of 5 kPa or firmer; a boat or low water to drive the stake	Medium	Bed strength and layering unknown at every site; placing the stake is the most exposed job	BKH-STM-001 S17; BKH-RSK-001 H1
Users	One person hauls; two lift the stake	Medium	Pull strength of the users and their acceptance of loop-set nets not yet known	BKH-STM-001 S4, S26

Ratings: High, Medium, Low.